

BXT CIE Unit 4

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Batch: B

Task 0

Name	Description
CircuitJS (Falstad)	A browser based simulator to quickly build and test electronic circuits.
Tinkercad	An online platform by Autodesk for beginners to design 3D electronics and Arduino models.
Multism	A professional circuit design and simulation software for detailed electronic circuits.

From these four tools, Falstad was selected. It is an online website made to build and test electronic circuits for beginners.

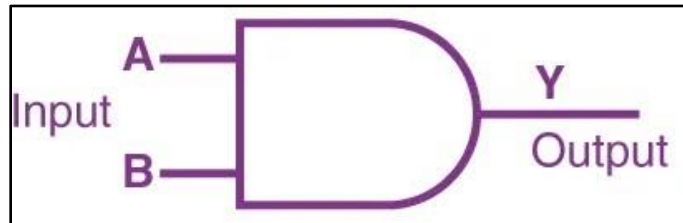
This website was chosen because:

1. It is a free website which requires no software download unlike multism.
2. The various different types of electronic parts are very easy to find making it very user friendly.
3. It contains all the logic gates and other components required for this assignment.

Task 1

AND Gate:

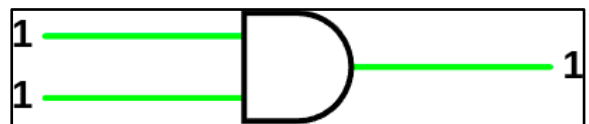
1. Circuit Diagram:



2. Truth Table:

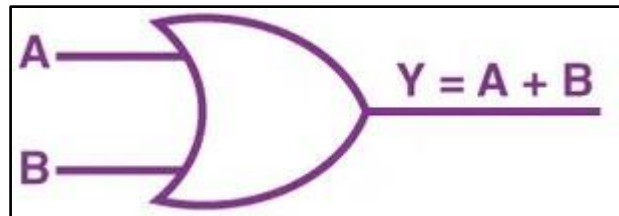
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

3. Circuit on Falstad:



OR Gate:

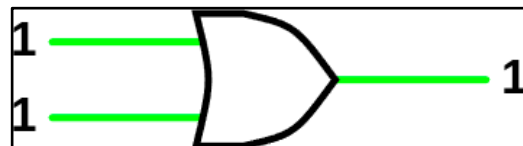
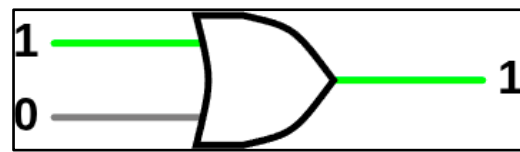
1. Circuit Diagram:



2. Truth Table:

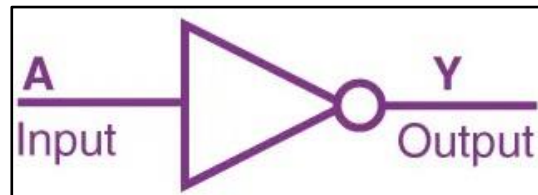
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

3. Circuit on Falstad:



NOT Gate:

1. Circuit Diagram:



2. Truth Table:

A	Output
0	1
1	0

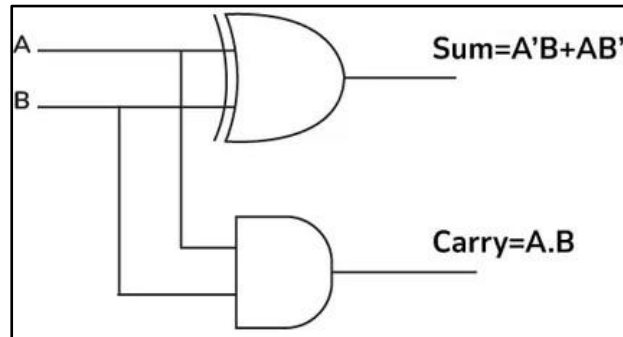
3. Circuit on Falstad:



Task 2

Half Adder:

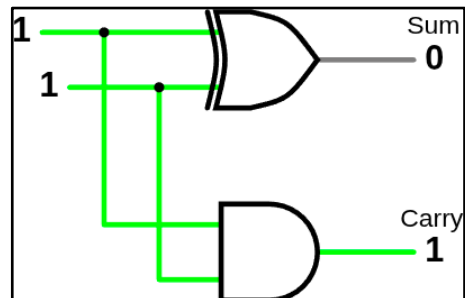
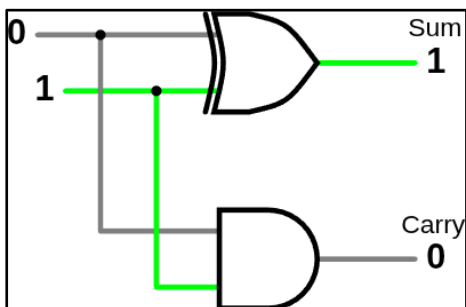
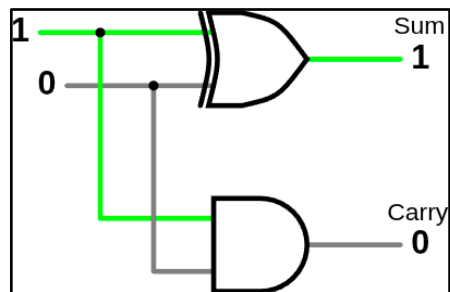
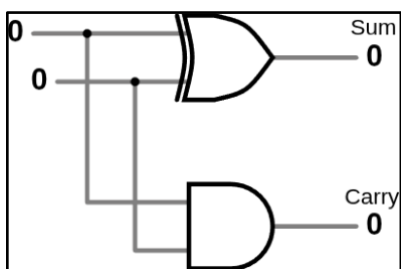
1. Circuit Diagram:



2. Truth Table:

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

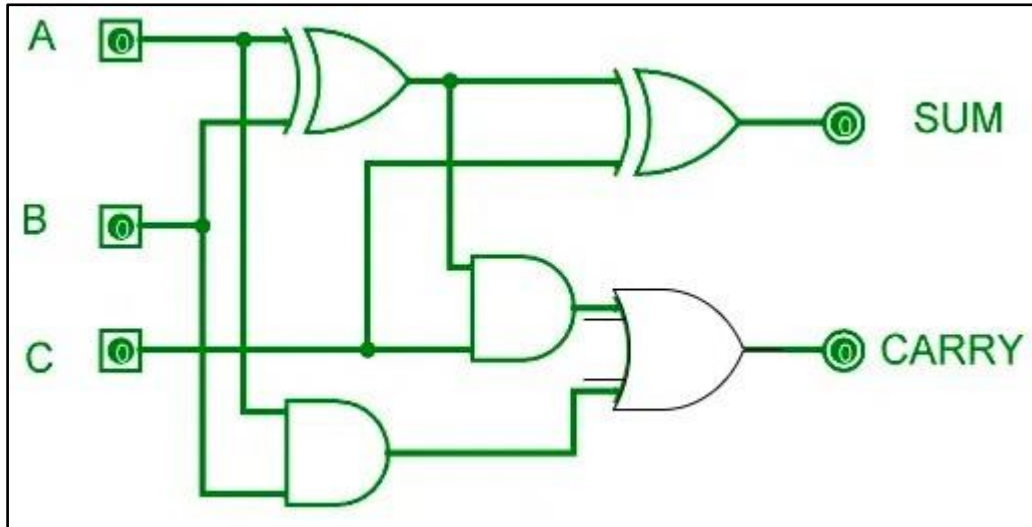
3. Circuit on Falstad:



Task 3

Full Adder:

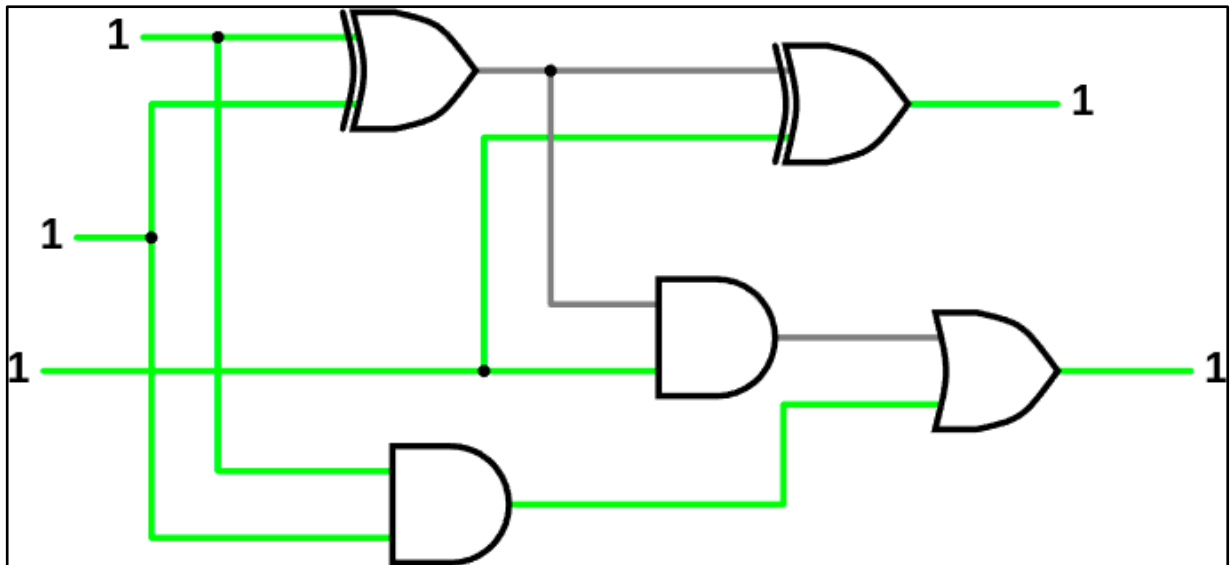
1. Circuit Diagram:



2. Truth Table:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

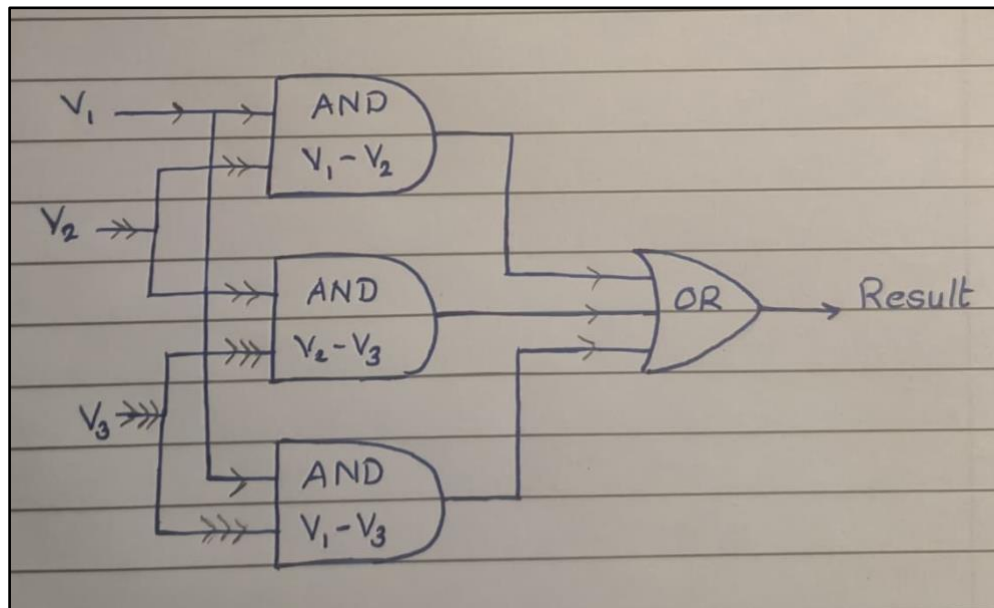
3. Circuit on Falstad:



Task 4

Digital Voting System

1. Circuit Diagram:



2. Explanation:

- 3 votes cast YES (1) or NO (0). The motion passes only if at least 2 out of 3 vote YES.

- b. Each AND gate checks one pair of voters. If any two people agree on YES, that AND gate outputs 1. The final OR gate checks if any pair agreed, if yes, the final output is YES.
- c. Boolean Expression: $\text{Result} = (V1.V2) + (V2.V3) + (V1.V3)$
- d. This can be used in the real world for voting machines or while voting on decisions in a group.

3. Table:

V1	V2	V3	Result
0	0	0	Rejected
0	0	1	Rejected
0	1	0	Rejected
0	1	1	Passed
1	0	0	Rejected
1	0	1	Passed
1	1	0	Passed
1	1	1	Passed

4. Circuit on Falstad:

